Adaptive Deep Learning and Machine Learning Model Selection in Sketch-to-3D Face Reconstruction for Forensic Applications

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Abstract*—Forensic facial reconstruction from hand-drawn sketches plays a pivotal role in criminal identification. However, single-model architectures often struggle with domain gaps, varying artist sketch styles, and structural degradation in sketches. This paper presents an end-to-end, web-based automated framework that converts forensic hand-drawn sketches into realistic facial reconstructions with an interactive visualization layer. The system trains and evaluates eight Machine Learning and Deep Learning architectures — Random Forest, XGBoost, and K-Nearest Neighbors from classical ML, and a convolutional autoencoder (CNN), a pretrained ResNet50 encoder-decoder, a UNet-style deep autoencoder, an unconditional GAN, and a Pix2Pix-style conditional GAN (cGAN) from deep learning — under a two-stage evaluation and selection protocol. In Stage 1, all eight models are ranked on a held-out validation split using a composite score combining SSIM, PSNR, F1 (derived from a per-pixel reconstruction-tolerance classification), and inverse MSE, identifying the strongest model overall. In Stage 2, because no ground-truth photo exists for a genuinely new forensic sketch at inference time — making reference-based metrics uncomputable — the system applies a dynamic, reference-free selection engine that scores every model's output using no-reference image-quality signals (sharpness, entropy, edge density, contrast, local-texture variance, and face-detection confidence) and automatically routes to the highest-scoring reconstruction for that specific sketch. The selected 2D output is projected onto a head-proportioned 3D mesh and rendered in-browser via Three.js/WebGL with full rotation, zoom, pan, and GLB export, providing an interactive visualization layer without requiring specialist computer-vision tooling. We report complete, measured results for all eight models on a real sketch/photo dataset (2,880 pairs), describe the system architecture and mathematical formulation of each model and the two-stage selection procedure, and discuss the classical-versus-deep and adversarial-versus-pixel-fidelity trade-offs observed. True geometric 3D face reconstruction (e.g., via a 3D Morphable Model / DECA) is identified as a concrete, scoped direction for future work.

***Index Terms*—Forensic Facial Reconstruction, Multi-Model Selection, Generative Adversarial Networks, Deep Autoencoders, No-Reference Image Quality Assessment, WebGL, Three.js.**

I. INTRODUCTION

Forensic facial sketches drawn by eyewitness identification experts serve as key preliminary intelligence in criminal investigations. Translating these highly qualitative, sparse, and textureless hand-drawn sketches into photorealistic 2D facial depictions and an interactive visual render remains a major challenge in biometrics and computer vision.

Existing solutions primarily deploy a single static deep generative framework — such as a conditional GAN — to perform sketch-to-photo translation [2]. However, these paradigms can struggle when faced with non-linear artistic noise, varying sketch strokes, missing structural proportions, and domain bias across different populations [1]. A single architecture is unlikely to perform optimally across all forensic sketch variations, since the 'best' generator is plausibly a function of input sketch complexity, stroke density, and missing-region severity — properties that vary sketch-to-sketch rather than dataset-to-dataset.

To address this, we implement and empirically compare an automated multi-model selection pipeline integrated into a full working web application. The key contributions of this paper are as follows:

Heterogeneous Model Pipeline: Full implementation, training, and empirical comparison of eight diverse architectures spanning classical ML (Random Forest, XGBoost, K-Nearest Neighbors) and deep learning (CNN autoencoder, ResNet50 encoder-decoder, UNet-style deep autoencoder, unconditional GAN, Pix2Pix-style [2] cGAN), with complete measured results on a real 2,880-pair sketch/photo dataset.

Two-Stage Automated Model Selection: a validation-set composite ranking (Stage 1) identifying the strongest model overall, and a genuinely dynamic, per-sketch, reference-free selection engine (Stage 2, Sec. III-C) scoring each candidate using no-reference quality signals — required because no ground truth exists for a new forensic sketch.

Interactive Web-Based Visualization: a Three.js/WebGL viewer projecting the selected 2D reconstruction onto a head-proportioned 3D mesh, with full manipulation and GLB export — explicitly distinguished from true geometric 3D reconstruction (Sec. III-D).

Reproducible, Fully-Measured Evaluation: complete training and validation results for all eight models (Sec. IV), not a partial or projected benchmark.

II. RELATED WORK

Traditional sketch-to-photo conversion relied heavily on subspace learning, such as Eigenfaces and locally linear embedding (LLE) [1]. These methods struggle to generalize non-linear stroke distortions. Modern advancements leverage deep convolutional neural networks (CNNs) and generative adversarial networks (GANs). Isola et al. introduced Pix2Pix for image-to-image translation using paired adversarial loss [2] — the architecture family our conditional GAN (Sec. III-B7) follows — while Zhu et al. achieved unpaired translation using cycle-consistency constraints in CycleGAN [3]. Ronneberger et al. proposed the U-Net encoder-decoder with skip connections, originally for biomedical segmentation but widely adopted for dense image translation [6]; our deep autoencoder (Sec. III-B6) follows this design. Recent diffusion models, notably Stable Diffusion [4] fine-tuned with ControlNet-style edge conditioning [8], have set new benchmarks in conditional synthesis and represent a natural, higher-compute extension of the present pipeline (Sec. VI). Perceptual similarity correlates more closely with human judgment than pixel-wise metrics through LPIPS [7], while Fréchet Inception Distance (FID) [9] assesses distributional realism; we discuss a perceptual (VGG feature-space) loss term, following Johnson et al. [13], in Sec. IV-C. On the 3D side, Feng et al.'s DECA framework [5] recovers detailed, animatable 3D face geometry by regressing 3DMM (FLAME [10]) parameters; Sec. III-D discusses the concrete, hardware-specific reasons true geometric reconstruction was not attempted here. Ensemble tree methods such as Random Forest [11] and gradient-boosted trees (XGBoost) [12] remain relevant as lightweight baselines, and both are fully implemented and evaluated in this work.

Existing literature rarely addresses real-world deployment realities: single-model fragility across heterogeneous sketch inputs, the absence of ground-truth references at true deployment time, and the lack of an accessible visualization layer for forensic personnel who are typically not computer-vision specialists. The present work implements and measures a complete system addressing each point directly.

III. PROPOSED SYSTEM ARCHITECTURE

The end-to-end framework consists of four coupled subsystems: Preprocessing, Multi-Model Inference, Two-Stage Evaluation & Selection, and Interactive Web Visualization. Section V describes the concrete deployment architecture.

A. Preprocessing Module

Upon sketch upload, the pipeline resizes the input, converts to grayscale, applies median-blur denoising, enhances local contrast via CLAHE, applies histogram equalization, and performs Haar-cascade face alignment — skipped gracefully when no face is detected — before normalizing:

$$I_{norm} = (I_{raw} - \mu_I) / (\sigma_I + \varepsilon) \quad (1)$$

where I_{raw} is the input sketch tensor, μ_I and σ_I are the channel-wise mean and standard deviation, and ε is a small stability constant.

B. Multi-Model Inference Pipeline

The normalized tensor I_{norm} is dispatched to eight independently trained architectures sharing a common train/validation split. We formalize each below.

1) Random Forest (RF) Regression: The sketch is reduced to a feature vector x via HOG and statistical descriptors. An ensemble of T regression trees, one per output dimension, predicts a flattened, downsampled target vector:

$$\blacksquare_j = (1/T) \Sigma\, h_t(x) \text{ per output dim. } j \quad (2)$$

2) XGBoost Objective: minimizes a regularized, second-order Taylor-approximated objective over an additive ensemble of K trees f_k:

$$L \approx \Sigma\, [g_i f_t(x_i) + (1/2)\, h_i f_t^2(x_i)] + \Omega(f_t) \quad (3)$$

$$\Omega(f_t) = \gamma T + (1/2)\, \lambda\, \Sigma\, w_j^2 \quad (4)$$

3) K-Nearest Neighbors (KNN): distance-weighted average of the k nearest training targets:

$$\blacksquare = \Sigma\, (1/d(x, x_i))\, y_i \,/\, \Sigma\, (1/d(x, x_i)) \quad (5)$$

Unlike RF/XGBoost, KNN natively supports multi-output regression in one call.

4) CNN Autoencoder: plain conv. encoder-decoder, no skip connections, L1 loss:

$$L_{CNN} = E_{x,y}\, ||y - G(x)||_1 \quad (6)$$

5) ResNet50 Encoder-Decoder: pretrained ImageNet ResNet50 encoder, conv. decoder, same L1 objective as (6).

6) Deep Autoencoder (UNet-style): encodes through L downsampling blocks, reconstructs via a symmetric expanding path with skip connections:

$$d_l = U_l[d_{l+1}\uparrow;\, e_l],\ l = L-1, \ldots, 1 \quad (7)$$

Skip connections preserve high-frequency detail (eye/nose contours). Trained with L1 loss only, isolating architectural effects from adversarial loss.

7) GAN (Unconditional Discriminator): the same UNet generator trained against a discriminator judging only realism:

$$L_{GAN}(G,D) = E_y[\log D(y)] + E_x[\log(1 - D(G(x)))] \quad (8)$$

$$G^* = \text{argmin}_G \max_D L_{GAN}(G,D) + \lambda\, L_1(G) \quad (9)$$

8) Conditional GAN / Pix2Pix (cGAN): same generator, PatchGAN discriminator conditioned on the sketch [2]:

$$L_{cGAN}(G,D) = E_{x,y}[\log D(x,y)] + E_x[\log(1 - D(x,G(x)))] \quad (10)$$

$$G^* = \text{argmin}_G \max_D L_{cGAN}(G,D) + \lambda\, L_1(G) \quad (11)$$

C. Two-Stage Automated Model Selection

A central design goal is avoiding a hardcoded, always-selected 'best' model.

Stage 1 — Validation-Set Composite Ranking: after training, each model is scored on the held-out split:

$$S_1 = .35\, SSIM_{bar} + .25\, PSNR_{bar} + .25\, F1_{bar} + .15\, (1/MSE_{bar}) \quad (12)$$

each term min-max normalized across models. This yields the overall best-model designation (Table I).

Stage 2 — Dynamic, Reference-Free Selection: for a new sketch, no ground truth exists, so every model's output O_k is scored on signals computable from output alone:

$$S_2 = .22\, Sharp + .22\, Face + .16\, Edge + .14\, Entr + .13\, Tex + .13\, Cont \quad (13)$$

where Sharp is Laplacian variance, Entropy is Shannon entropy, Edge is Canny edge density, Contrast is intensity std. dev., Texture is local-variance, and Face is Haar-cascade detection confidence. The winner is routed: $O^* = \text{argmax}_k\, S_2(O_k)$ (14)

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Algorithm 1: Dynamic (Stage-2) Multi-Model Selection
Require: Preprocessed sketch I_norm; trained models {M1,...,M8}; weig
Ensure: Selected O*, index k*, ranked scores
1: scores ← ∅
2: for k = 1 to 8 do
3:   O_k ← M_k(I_norm)
4:   compute Sharp, Face, Edge, Entropy, Tex, Contrast
5:   S_2(O_k) ← weighted sum per (13)
6:   scores[k] ← S_2(O_k)
7: end for
8: k* ← argmax_k scores[k]
9: O* ← O_k*
10: return O*, k*, sorted(scores)
```

D. Interactive 3D Visualization

The selected 2D output O* is texture-mapped onto a parametric head-proportioned UV-sphere mesh. Full manipulation (360° rotation, zoom, pan) uses Three.js OrbitControls, with GLB export via THREE.GLTFExporter.

IV. EXPERIMENTS AND RESULTS

We evaluate on a real sketch/photo dataset of 2,880 paired images (85/15 train/val split). Table I reports complete measured results for all eight models.

TABLE I

PER-MODEL RECONSTRUCTION AND CLASSIFICATION METRICS

Model	Type	MSE	PSNR	SSIM	Accuracy	Precision	Recall	F1
Random Forest	classical	2680.9802	14.21	0.5435	0.393	0.995	0.357	0.515
XGBoost	classical	2286.8919	14.96	0.5641	0.446	0.970	0.427	0.581
K-Nearest Neighbors	classical	4031.4090	12.49	0.4602	0.354	0.994	0.316	0.470
CNN Autoencoder	deep	2224.2760	15.05	0.4199	0.479	0.939	0.486	0.631
Deep Autoencoder (best)	**deep**	**1458.9831**	**17.03**	**0.5639**	**0.571**	**0.936**	**0.591**	**0.714**
GAN	deep	1538.5053	16.75	0.5441	0.551	0.935	0.570	0.696
Conditional GAN (cGAN)	deep	1556.4223	16.70	0.5188	0.547	0.934	0.566	0.693
ResNet50 Encoder-Decoder	deep	1680.1693	16.32	0.4773	0.545	0.940	0.559	0.690

B. Two-Stage Selection & Dynamic Routing

Stage 1 ranks Deep Autoencoder as strongest overall (highest SSIM 0.5639, PSNR 17.03 dB, F1 0.714). Stage 2 dynamic engine routes per-sketch inputs to the optimal candidate based on no-reference quality signals.

C. Qualitative Analysis and Failure Modes

Deep models outperform classical regressors due to feature-flattening spatial detail loss. UNet-style Autoencoder (L1 only) outperforms GAN/cGAN on SSIM/PSNR since adversarial loss optimizes perceptual realism over exact pixel matching.

V. WEB APPLICATION SYSTEM DESIGN

Single-process Flask web app with server-rendered templates, in-memory model registry caching, JSON prediction history, and Three.js WebGL canvas.

VI. CONCLUSION AND FUTURE SCOPE

Fully implemented automated forensic sketch-to-face reconstruction system featuring two-stage model selection across eight architectures. Future work includes true 3DMM geometric face reconstruction (DECA/FLAME) and diffusion model integration.

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